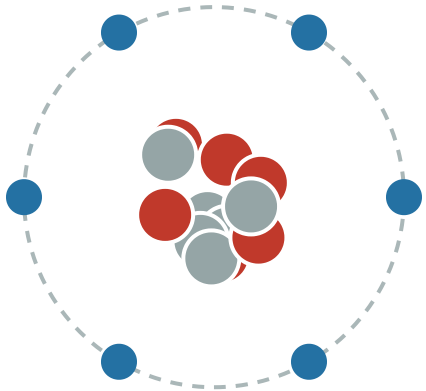
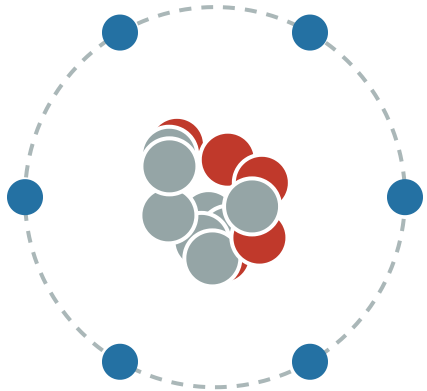
# Isotopes: same element, different number of neutrons

## Carbon-12 $^{12}_{6}C$



**6 protons** · **6 neutrons** · 6 electrons

## Carbon-14 $^{14}_{6}C$



**6 protons** · **8 neutrons** · 6 electrons

**Same Z = 6** → both are carbon · **Different A** → different mass

Carbon-14 is radioactive: it is used to find out the age of fossils and ancient remains